

# OBJECT ORIENTED PROGRAMMING

## LAB ASSIGNMENT –2

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### Problem 1:

In today's digital world, secure passwords are essential to protect sensitive data. Weak passwords (like only numbers or only alphabets) are highly vulnerable to cyberattacks. A password checker is required to verify the **strength of a password** based on its composition.

The program should read a password from the user and analyze its characters using a **for loop**. It must count uppercase letters, lowercase letters, digits, and special characters. Based on these counts, the program will determine if the password is **Weak, Moderate, or Strong**.

### Objectives:

- To use a **for loop** to traverse each character of the password string.
- To check and classify characters into:
  - Uppercase letters (A–Z)
  - Lowercase letters (a–z)
  - Digits (0–9)
  - Special characters (@, #, \$, %, !, etc.)
- To decide password strength based on composition rules.
- To demonstrate practical use of loops in real-world **cybersecurity applications**.

### Password Strength Rules:

- Weak:
  - Length < 6 OR contains only one type of character (e.g., only digits).
- Moderate:
  - Length ≥ 6 AND contains at least two types of characters (e.g., letters + digits).
- Strong:
  - Length ≥ 8 AND contains all four types (uppercase, lowercase, digits, special).

### Expected Outcome:

- Program should count and display character categories.
- Based on rules, display the password's strength.
- Should handle different test cases (weak, moderate, strong).

### SampleInput/Output

#### Input:

Enterpassword:abcd123

#### Output:

UppercaseLetters: 0

LowercaseLetters: 4

Digits:3

Special Characters: 0

PasswordStrength:Moderate

Solution :

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
    string pass, strength="Weak";
    int u=0, l=0, d=0, s=0;
    int U=0, L=0, D=0, S=0;
    cout << "\n\nPassword Strength Checker\n\nEnter Password : ";
    getline(cin, pass);
    int len=pass.length();
    for(char ch:pass)
    {
        if(ch >= 'a' && ch<='z')
            {l++;L=1;}

        else if(ch >= 'A' && ch<='Z')
            {u++;U=1;}

        else if(ch >= '0' && ch<='9')
            {d++;D=1;}

        else
            {s++;S=1;}
    }
    int sum=U+D+L+S;

    if(len >= 8 && sum==4 )
        strength="Strong";
    else if (len >=6 && sum>=2)
        strength = "Moderate";
}
```

```

    cout << endl << "\nUppercase Letters: " << u << "\nLowercase Letters: " << l <<
"\nDigits: " << d << "\nSpecial Characters: " << s << "\nPassword Strength: " <<
strength;
    return 0;
}

```

Output :

```

PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass2_Q1.exe"

Password Strength Checker

Enter Password : So this Would be my P44sword

Uppercase Letters: 3
Lowercase Letters: 18
Digits: 2
Special Characters: 5
Password Strength: Strong
PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass2_Q1.exe"

Password Strength Checker

Enter Password : 123456789

Uppercase Letters: 0
Lowercase Letters: 0
Digits: 9
Special Characters: 0
Password Strength: Weak
PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass2_Q1.exe"

Password Strength Checker

Enter Password : Meow broooo

Uppercase Letters: 1
Lowercase Letters: 9
Digits: 0
Special Characters: 1
Password Strength: Moderate

```

## Problem 2:

In real-world e-commerce or retail billing systems, customers purchase multiple items in a single transaction. The system must accept item prices one by one, calculate the total bill, and apply discounts if applicable. To simulate this scenario, we design a C++ program that repeatedly accepts item prices until the customer finishes shopping.

The loop should continue asking for prices until the user enters 0, which indicates checkout. Using break and continue statements ensures flexible control of the loop, similar to real-world billing systems.

**Objectives:**

- To simulate a shopping cart system where multiple item prices are entered.
- To use a for/while loop to repeatedly accept item prices.
- To apply:
  - continue → skip invalid entries (like negative prices).
  - break → stop when user enters 0 (checkout).
- To calculate the total bill and apply discounts if total exceeds a threshold.

**Discount Rule:**

- If total bill > ₹5000, apply 10% discount.
- Otherwise, no discount.

**Expected Outcome:**

- Program should allow the user to keep adding item prices.
- Negative prices should be ignored with a warning.
- Entering 0 should stop input and calculate the final bill.
- If applicable, discount should be applied automatically.

**SampleInput/Output****Input/Run:**

```
Enteritemprice(0tocheckout):1200
Enteritemprice(0tocheckout):1500 Enter
item price (0 to checkout): -50 Invalid
price! Skipped.
Enteritemprice(0tocheckout):3000 Enter
item price (0 to checkout): 0
```

**Output:**

```
TotalBill(beforediscount):₹5700
Discount Applied: ₹570
FinalBill:₹5130
```

Solution :

```
#include <iostream>

using namespace std ;

int main()
{
    float total = 0 , price, discount = 0 ;
    do
    {
        cout << endl << "Enter item price (0 to checkout) : " ;
        cin >> price ;
        if (price < 0)
        {
            cout << "\nInvalid price ! Skipped. \n";
            continue;
        }
        else if (price==0)
            break;
        else
            total += price ;
    } while( price != 0 );
    if (total > 5000)
        discount = total*0.1 ;
    cout << "\n\nTotal Bill (before discount): Rs. " << total << "\nDiscount
Applied: Rs. " << discount << "\nFinal Bill: " << total - discount ;
    return 0;
}
```

Output :

```
PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass2_Q2.exe"

Enter item price (0 to checkout) : 1000

Enter item price (0 to checkout) : 4000

Enter item price (0 to checkout) : 1

Enter item price (0 to checkout) : -699

Invalid price ! Skipped.

Enter item price (0 to checkout) : 0

Total Bill (before discount): Rs. 5001
Discount Applied: Rs. 500.1
Final Bill: 4500.9
```

**Problem 3:**

In real-world traffic management, signals are crucial to ensure road safety and regulate the flow of vehicles. Drivers must respond appropriately to the traffic light color. The problem is to design a program in C++ that simulates a traffic signal and provides instructions to the driver based on the signal color entered.

**Objective:**

- To develop a C++ program that takes the traffic signal color (Red, Yellow, or Green) as input.
- To implement decision control statements (if, else-if, else) for decision-making.
- To display the corresponding driving instruction:
  - Red → Stop
  - Yellow → Get Ready
  - Green → Go
- To handle invalid input using proper validation.

**Expected Outcome:**

- The program should correctly identify the input signal color and display the respective driving instruction.
- If the input is invalid (not Red, Yellow, or Green), the program should display an error message.
- The output should simulate a real-world traffic light system by guiding the driver's action.

**SampleInput/Output**

**Input:**

```
Entertrafficlightcolor:Red
```

**Output:**

```
Stop!
```

**Input:**

```
Entertrafficlightcolor:Green
```

**Output:**

```
Go!
```

**Input:**

```
Entertrafficlightcolor:Blue
```

**Output:**

```
Invalidinput
```

Code :

```
#include <iostream>
#include <string>
#include <cctype>

using namespace std ;

int main()
{
    string color,action;
    cout << endl << "Enter traffic light color : ";
    cin >> color ;
    for ( char &c:color)
    {
        c=tolower(c);
    }
    if (color=="red")
        cout << "STOP !";
    else if (color == "green")
        cout << "GO !";
    else if (color == "yellow")
        cout << "Get Ready...";
    else
        cout << "Invalid Input !";

    return 0;
}
```

Output :

```
PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass2_Q3.exe"

Enter traffic light color : red
STOP !
PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass2_Q3.exe"

Enter traffic light color : Green
GO !
PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass2_Q3.exe"

Enter traffic light color : YellOw
Get Ready...
PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass2_Q3.exe"

Enter traffic light color : Purple
Invalid Input !
```

----- THANK YOU -----